Math 190-1: Mathematics of Sustainability **Spring 2026**

Lecture 14: Probability and Markov Chains

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14.1 Probability Overview

Definition 14.1. *The term* experiment *is used to refer to any process whose outcome is not known in advance, such as flipping a coin or rolling a die. The* sample space *associated with an experiment is the set of all possible outcomes. The sample space is usually denoted by* Ω*, the capital Greek letter Omega*

Example 14.2. *The flip of one coin has two possible outcomes, called "Heads" and "Tails," and denoted by* H *and* T*. Flipping three coins leads to* $2^3 = 8$ *outcomes:*

$$\begin{array}{cccc} & HHT & HTT & \\ HHH & HTH & THT & TTT \\ & THH & TTH & \end{array}$$

Example 14.3. *The roll of one die has six possible outcomes: 1, 2, 3, 4, 5, 6. Rolling two dice leads to* $6^2 = 36$ *outcomes* $\{(m, n) : 1 \leq m, n \leq 6\}$.

Definition 14.4. *Intuitively, an* event *is a statement about the outcome of an experiment. Formally, an event is a subset of the sample space.*

Example 14.5. *An example for flipping three coins is "two coins show Heads," or*

$$A = \{HHT, HTH, THH\}$$

An example for rolling two dice is "the sum is 9," or

$$B = \{(6, 3), (5, 4), (4, 5), (3, 6)\}$$

Definition 14.6. *Events are just sets, so we can perform the usual operations of set theory on them. For example, if* $\Omega = \{1, 2, 3, 4, 5, 6\}, A = \{1, 2, 3\}$*, and* $B = \{2, 3, 4, 5\}$*, then the* union $A \cup B = \{1, 2, 3, 4, 5\}$*, the* intersection $A \cap B = \{2, 3\}$*, and the* complement of $A, A^c = \{4, 5, 6\}$*. To introduce our next definition, we need one more notion: two events are* disjoint *if their intersection is the empty set,* $\emptyset$*.* A *and* B *are not disjoint, but if* $C = \{5, 6\}$*, then* A *and* C *are disjoint.*

Definition 14.7. *A* probability *is a way of assigning numbers to events that satisfies:*

1. *For any event* $A, 0 \leq P(A) \leq 1$.
2. *If* Ω *is the sample space, then* $P(\Omega) = 1$.
3. *the probability of a union of disjoint events is the sum of the proba- bilities of the sets.*

$$P\left(\bigcup_{i=1}^{k} A_i\right) = \sum_{i=1}^{k} P(A_i)$$

Example 14.8. *For a very general example of a probability, let $\Omega = \{1, 2, ..., n\}$; let $p_i \leq 0$ with $\Sigma_i p_i = 1$; and define $P(A) = \Sigma_i p_i$ for all i in A. Two basic properties that follow immediately from the definition of a probability are*

$$P(A) = 1 - P(A^c)$$
$$P(B \cup C) = P(B) + P(C) - P(B \cap C)$$

Example 14.9. *Roll two dice and suppose for simplicity that they are red and green. Let A = "at least one 4 appears," B = "a 4 appears on the red die," and C = "a 4 appears on the green die," so $A = B \cup C$. Find the Probability of A, $P(A)$ two different ways.*

First, A^c = "neither die shows a 4," which contains $5 \cdot 5 = 25$ outcomes so $P(A) = 1 - 25/36 = 11/36$.

Secondly, $A = B \cup C$, so $P(B) = P(C) = 1/6$ while $P(B \cap C) = P(\{4, 4\}) = 1/36$, so $P(A) = 1/6 + 1/6 - 1/36 = 11/36$.

Definition 14.10. *Suppose we are told that the event A with $P(A) > 0$ occurs. Then the sample space is reduced from Ω to A and the probability that B will occur given that A has occurred is*

$$P(B|A) = P(B \cap A)/P(A).$$

To explain this formula, note that only the part of B that lies in A can possibly occur, and since the sample space is now A, we have to divide by $P(A)$ to make $P(A|A) = 1$. Multiplying on each side by $P(A)$:

$$P(B|A)P(A) = P(B \cap A).$$

Example 14.11. *Suppose we draw two balls without replacement from an urn with 6 blue balls and 4 red balls. What is the probability we will get two blue balls? Let A = blue on the first draw, and B = blue on the second draw. Clearly, $P(A) = 6/10$. After A occurs, the urn has 5 blue balls and 4 red balls, so $P(B|A) = 5/9$ and it follows that*

$$P(A \cap B) = P(A)P(B|A) = \frac{6}{10} \cdot \frac{5}{9}$$

To see that this is the right answer notice that if we draw two balls without replacement and keep track of the order of the draws, then there are $10 \cdot 9$ outcomes, while $6 \cdot 5$ of these result in two blue balls being drawn.

Definition 14.12. *Last but far from least, two events A and B are said to be* independent *if $P(B|A) = P(B)$. In words, knowing that A occurs does not change the probability that B occurs. Using the multiplication rule this definition can be written in a more symmetric way as*

$$P(A \cap B) = P(A)P(B)$$

Definition 14.13. *There are two ways of extending the definition of independence to more than two events. $A1_, ..., A_n$ are said to be* pairwise independent *if for each $i \neq j, P(A_i \cap A_j) = P(A_i)P(A_j)$, that is, each pair is independent. $A_1, ..., A_n$ are said to be* independent *if for any $1 \leq i_1 < i_2 < ... < i_k \leq n$, we have*

$$P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$$

14.2 Markov Chains

The importance of Markov chains comes from two facts: (i) there are a large number of physical, biological, economic, and social phenomena that can be modeled in this way, and (ii) there is a well-developed theory that allows us to do computations.

Figure 14.1:

Example 14.14.

Example 14.15. *Consider a gambling game in which on any turn you win \$1 with probability $p = 0.4$ or lose \$1 with probability $1 - p = 0.6$. Suppose further that you adopt the rule that you quit playing if your fortune reaches \$N . Of course, if your fortune reaches \$0 the casino makes you stop.*

Definition 14.16. *Let X_n be the amount of money you have after n plays. Your fortune, X_n has the "Markov property." In words, this means that given the current state, X_n, any other information about the past is irrelevant for predicting the next state X_{n+1}.*

Definition 14.17. *We say that X_n is a discrete time* Markov chain *with* transition matrix $p(i, j)$ *if*

$$P(X_{n+1} = j|X_n = i) = p(i, j)$$

Example 14.18. *Describing the Random Walk in terms of a probability matrix.*

Example 14.19. *Describing the Gambler's ruin in terms of a probability matrix.*

$$\begin{aligned} p(0,0) &= 1 \\ p(N,N) &= 1 \\ p(i,i+1) &= 0.4, \quad 0 < i < N \\ p(i,i-1) &= 0.6, \quad 0 < i < N \end{aligned}$$

if we make a matrix out of this we have

We can also view this as a collection of states and ways to move between them.

	0	**1**	**2**	**3**	**4**	**5**
0	1.0	0	0	0	0	0
1	0.6	0	0.4	0	0	0
2	0	0.6	0	0.4	0	0
3	0	0	0.6	0	0.4	0
4	0	0	0	0.6	0	0.4
5	0	0	0	0	0	1.0

$$\underset{1}{\hookrightarrow} \quad 0 \quad \underset{0.6}{\longleftarrow} \quad 1 \quad \underset{0.6}{\overset{0.4}{\rightleftharpoons}} \quad 2 \quad \underset{0.6}{\overset{0.4}{\rightleftharpoons}} \quad 3 \quad \underset{0.6}{\overset{0.4}{\rightleftharpoons}} \quad 4 \quad \overset{0.4}{\longrightarrow} \quad 5 \quad \overset{1}{\hookleftarrow}$$